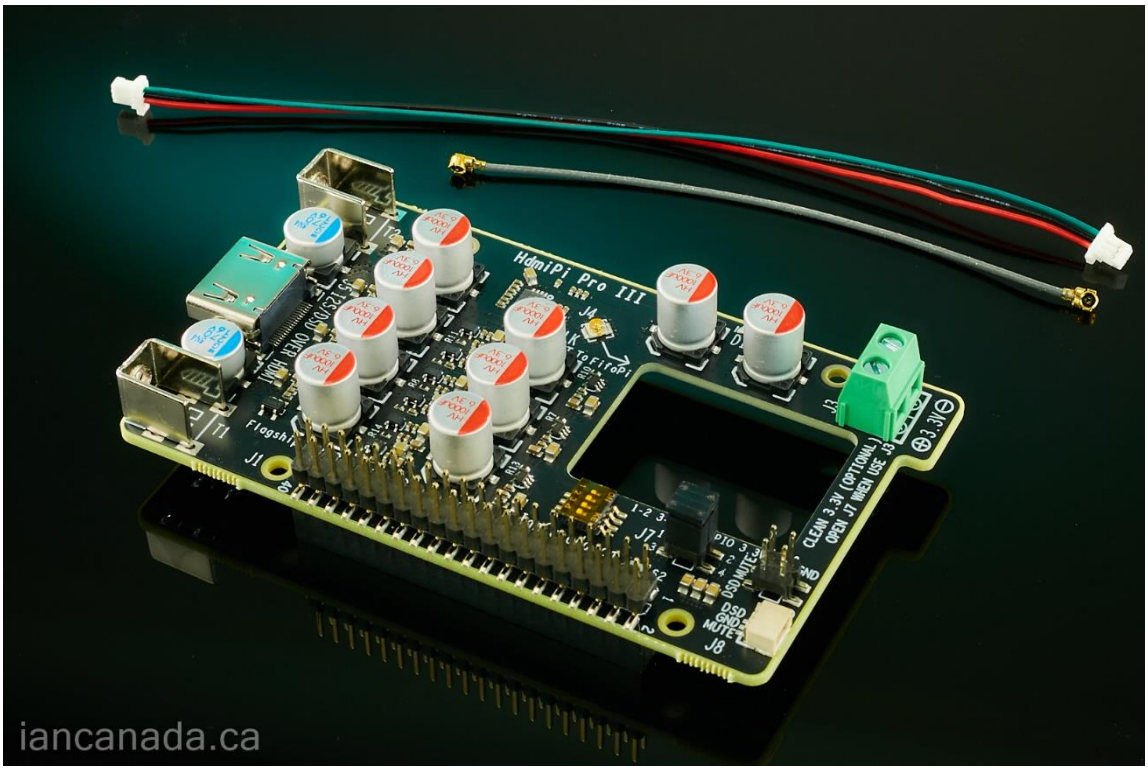
HdmiPi Pro

HdmiPi Pro II

HdmiPi Pro III

Flagship Ultra-low jitter I2S/DSD over HDMI transport interface

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A. Introduction

HdmiPi Pro is a dedicated I2S/DSD over HDMI transport interface board. Compared with all other solutions, it is the first one that uses discrete 1-bit LVDS drivers to achieve the highest HDMI signal quality and lowest noise levels. And it also integrates the well-reputed 1-bit discrete secondary re-clockers (similar to a ReClockPi) to ensure the best possible low-jitter performance. HdmiPi Pro is so far the flagship HDMI transport interface board for audiophiles to improve the sound quality of an external DAC to a higher level.

HdmiPi Pro works in SYNC clock mode so it will need a FifoPiMa, FifoPiQ7 or Q3 to operate.

B. Highlighted Features and Specifications

- Discrete 1-bit LVDS drives to achieve the best possible LVDS signal quality and the lowest noise levels
- Built-in the well-reputed 1-bit discrete secondary re-clock stage working in SYNC mode
- Enhanced high bandwidth power supply filtering networks
- LDO free design to make it directly use the 3.3V ultracapacitor or battery power supply for the best power supply performance
- Could be so far the best I2S/DSD over HDMI streamer transport interface
- Optimized PCB impedance control for the high speed LVDS signals
- 4 layers PCB design with dedicate shield layers to eliminate EMI noise
- Metal film ultra-low noise resistors to ensure the lowest possible noise level
- Pure sync mode works with FifoPiQ7, FifoPiMa or FifoPiQ3 or other re-clockers.
- Capable of I2S music up to 1536KHz
- Capable of native DSD music up to DSD1024

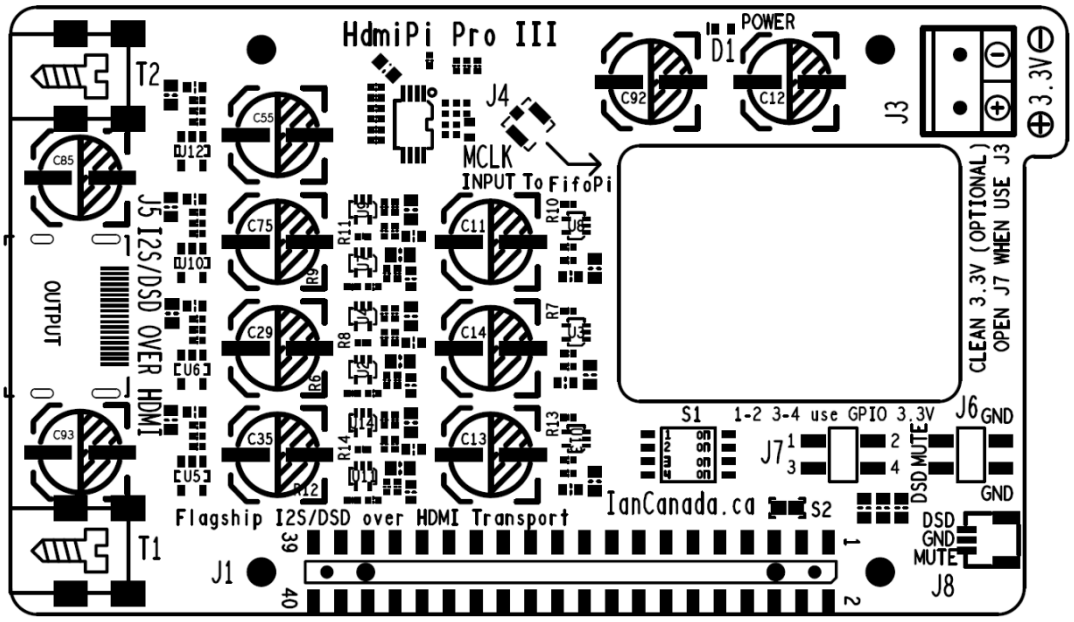
C. HdmiPi Pro II New Features

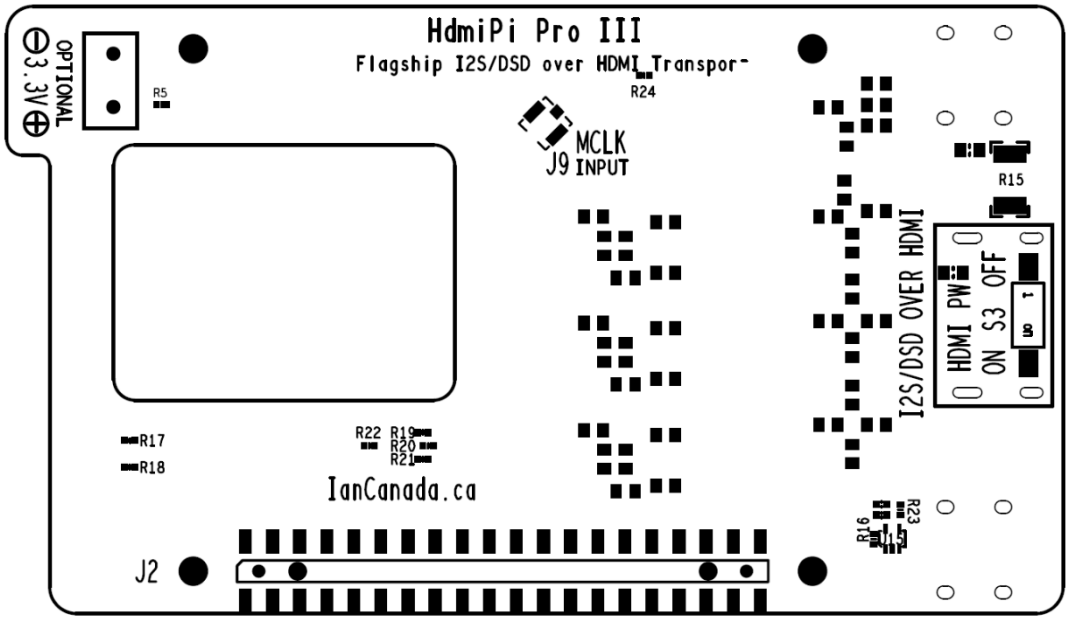
- Configurable HDMI format setting switches, works with all I2S/DSD over HDMI input DACs, even they are not compatible with the standard PS Audio format
- Optimized re-clocking scheme
- Optimized power local supply decoupling network
- Dedicated MUTE/DSDEN control connector that can be connected directly to the new FifoPiQ7II
- Mounting bracket for the HDMI connector

D. HdmiPi Pro III New Features and Improvements

- **New Low-Jitter Clock Drivers.** Equipped with newly optimized clock drivers that achieve approximately 50% lower additive phase jitter compared with the previous version, providing the potential for further improvements in overall audio performance.
- **Optimized for Full Fs Frequency Range.** The signal path has been carefully optimized for high-speed signal integrity across the full supported Fs range, such as PCM 768 kHz and DSD1024. This ensures reliable signal transmission and stable operation even under the most demanding high-resolution audio conditions.
- **Additional U.FL MCLK Input.** An additional U.FL MCLK input socket is located on the bottom side of the PCB, allowing the use of an ultra-short 2-inch U.FL cable. This makes it possible to minimize the MCLK signal path and potentially achieve further clock performance improvements.
- **Enhanced Local Power Supply Filtering.** Enhanced local power-supply filtering networks feature ultra-low-ESR capacitors and high-bandwidth filtering to achieve the lowest possible power-supply noise floor and provide cleaner power to critical circuits.
- **Selectable HDMI Power Supply.** The HdmiPiPro III features providing power supply to the HDMI receiver directly from the HDMI cable. This feature can be enabled or disabled using the S3 jumper switch, providing greater flexibility for different system configurations.
- **Larger PCB with Improved Silkscreen.** A slightly larger PCB provides additional layout flexibility, while improved silkscreen markings enhance legibility and ease of installation and configuration.
- **Precision Manufacturing.** Manufactured with precision engineering and careful attention to detail to ensure a high level of hardware quality and consistency.
- **New Low-Jitter Clock Drivers.** Equipped with newly optimized clock drivers that achieve approximately 50% lower additive phase jitter compared with the previous version, providing the potential for further improvements in overall audio performance.

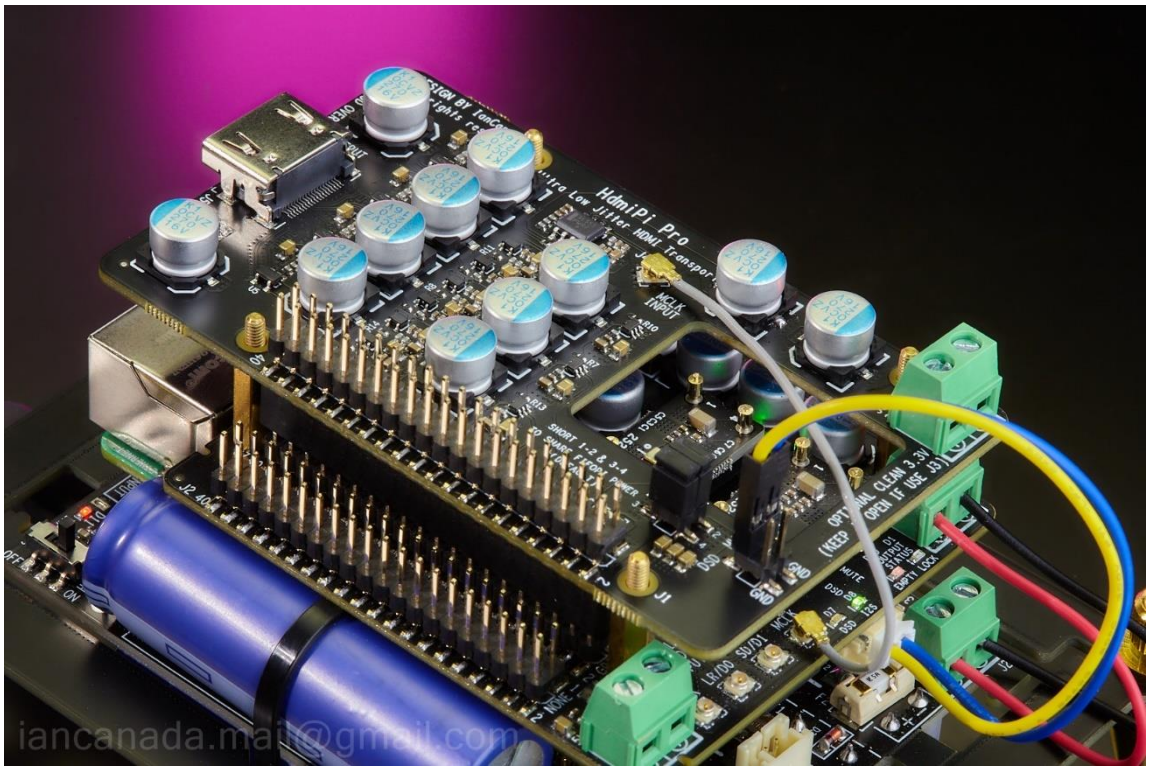
E. Layout


HdmiPi Pro III
POWER
D1
J3
3.3V
T2
T1
J4
MCLK
INPUT To FifoPi
J5 I2S/DSD OVER HDMI
OUTPUT
CLEAN 3.3V (OPTIONAL)
OPEN J7 WHEN USE J3
S1
1-2 3-4 use GPIO 3.3V
J6
GND
J7
DSD MUTE
GND
IanCanada.ca
S2
DSD
GND
MUTE
J8
Flagship I2S/DSD over HDMI Transport
J1
39
40
1
2


HdmiPi Pro III
Flagship I2S/DSD over HDMI Transport
OPTIONAL
3.3V
R5
R24
J9
MCLK
INPUT
R15
I2S/DSD OVER HDMI
HDMI PW
ON S3 OFF
R17
R18
R22 R19
R20
R21
IanCanada.ca
R16
R23
J2

F. Getting start

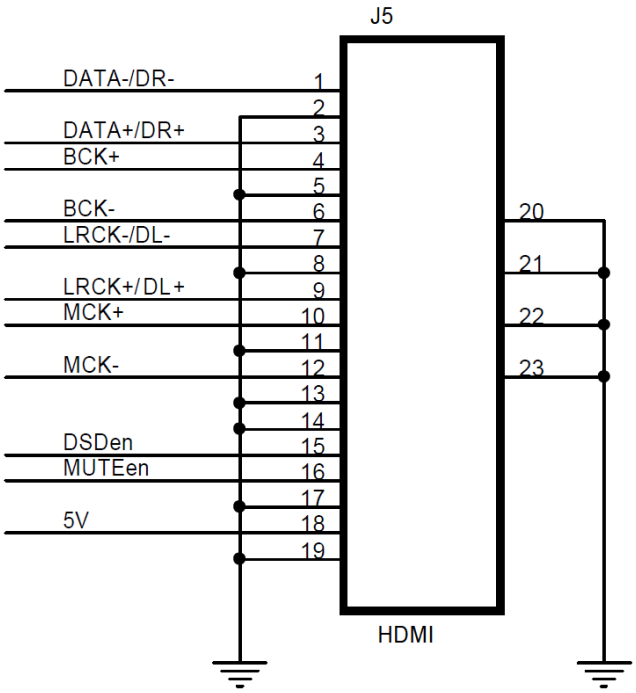
1. Make sure you FifoPi and rest of the system are configured well and working properly.
2. Install the HdmiPi Pro into the GPIO port of the FifoPi. Please use the 13mm standoffs.
3. Connect the MCLK from FifoPi to the J4 of HdmiPi Pro by the supplied U.FL coaxial cable. Connect the DSD and MUTE signals (optional) from FifoPi to J6 of the HdmiPi Pro if your external DAC takes those control signal through the HDMI cable. Don't need to connect the GND.
4. Make sure your external DAC's HDMI interface configurations are match the format of the HdmiPi's output signals. Connect a HDMI cable (4K, V2.0 or higher) from the HdmiPi Pro output connector J5 to the HDMI input of the external DAC.
5. Turn on the power supply. Make sure POWER LED D1 is light up. Play the music files as normal. Enjoy the music.



G. Connectors and setting switch

J5: HDMI output connector

Standard HDMI type A connector. To output LVDS PCM/DSD/DoP signals to receiver through HDMI cable.
For higher signal quality, high speed HDMI cables version 2.0 (4K) or higher are recommended.
Please refer the following schematic for the default HDMI configuration (PS Audio format).



S1: HDMI format configuration setting switch (HdmiPi Pro II, III)

Default settings: SW1, SW2, SW3, SW4 all off. For PS Audio format.

SCK/BCK phase	HDMI 4,6 configuration	
SW1	Pin-4	pin-6
OFF (*default)	BCK+	BCK-
ON	BCK-	BCK+

LRCK/DL phase	HDMI 7,9 configuration	
SW2	Pin-7	pin-9
OFF (*default)	LRCK-/DL-	LRCK+/DL+
ON	LRCK+/DL+	LRCK-/DL-

DATA/DR phase	HDMI 1,3 configuration	
SW3	Pin-1	pin-3
OFF (*default)	DATA-/DR-	DATA+/DR+
ON	DATA+/DR+	DATA-/DR-

MUTE phase	HDMI 16 configuration	
SW4	Pin-16	
OFF (*default)	MUTE input=1: pin16=0	MUTE input =0: pin16=1
ON	MUTE input=1: pin16=1	MUTE input =0: pin16=0

Note: MUTE input is the MUTE input signal on J6 and J8

J1, J2: 40 pin GPIO connectors

pin number	**J2** 40 PIN GPIO connector to board below (Raspberry Pi, IsolatorPi I/II, FiFoPi, or similar)	**J1** Optional 40 PIN GPIO connector to HAT on top
1,17	3.3V from preceding board	3.3V from preceding board
2,4	5V or 3.3V from preceding board	5V or 3.3V from preceding board
6,9,14,20, 25,30,34, 39	GND	GND
12	SCK input	SCK from preceding board
35	LRCK/DL input	LRCK/DL PIN from preceding board
40	DATA/DR input	DATA/DR PIN from preceding board
All other pins	same pin from preceding board	same pin from preceding board

40 pin GPIO connector note: All input/output signals are LVTTL (3.3V) logic level except power and ground.

J6, J8 (HdmiPi Pro II only): DSD and MUTE control signals input

HDMI Control	Low or unconnected	High
DSD	To play PCM	To play DSD
MUTE	To play normally	To mute DAC

Note: GND signals on J6 don't need to be connected to FifoPi

Note: J6 and J8 are equivalent. J8 is designed to connected to the new FifoPiQ7II directly through the supplied 3P SH1.0 cable.

J4, J9 (HdmiPi Pro III only): External MCLK input (u.fl coaxial cable socket)

When including the MCLK signal into HDMI cable, connect the external MCLK signal from a FifoPi or similar low jitter re-clocking source using a u.fl coaxial cable.

J3: Optional 3.3V DC power supply input

By default, HdmiPi Pro takes 3.3V power supply from FifoPi clean side through GPIO. If you want an independent power supply for HdmiPi Pro, you can connect a 3.3V (100mA or higher) power supply to the 2-pin 5.0mm terminal J3. MAINTAINING CORRECT POLARITY!!! J7 jumpers must keep open in this case. Low noise linear 3.3V power supply will be good for HdmiPi Pro. 3.3V ultracapacitor or LifePO4 battery power supplies are great and highly recommended.

J7: Power supply jumpers

Jumpers	**Short**	**Open**
1-2 3-4	Takes 3.3V power supply from FifoPi through GPIO	Takes 3.3V power supply from J3

S3 (HdmiPiPro III only): HDMI cable power supply enable/disable

ON: Enable HDMI cable 3.3V power supply on Pin18 of J5

OFF (1): Disable HDMI cable 3.3V power supply for Pin18 of J5 (default)

H. LED indicator

D3: Power indicator, Light up to indicate that the 3.3V power supply is applied.

I. Application examples

1. Compact size Ultra-low jitter Raspberry Pi I2S/DSD over HDMI transport

Components

(1). HdmiPi Pro

(2). FifiPi Q7 (or FifoPi Q3, or FifoPi Ma)

(3). MonitorPi (optional)

(4). Raspberry Pi

Power supply

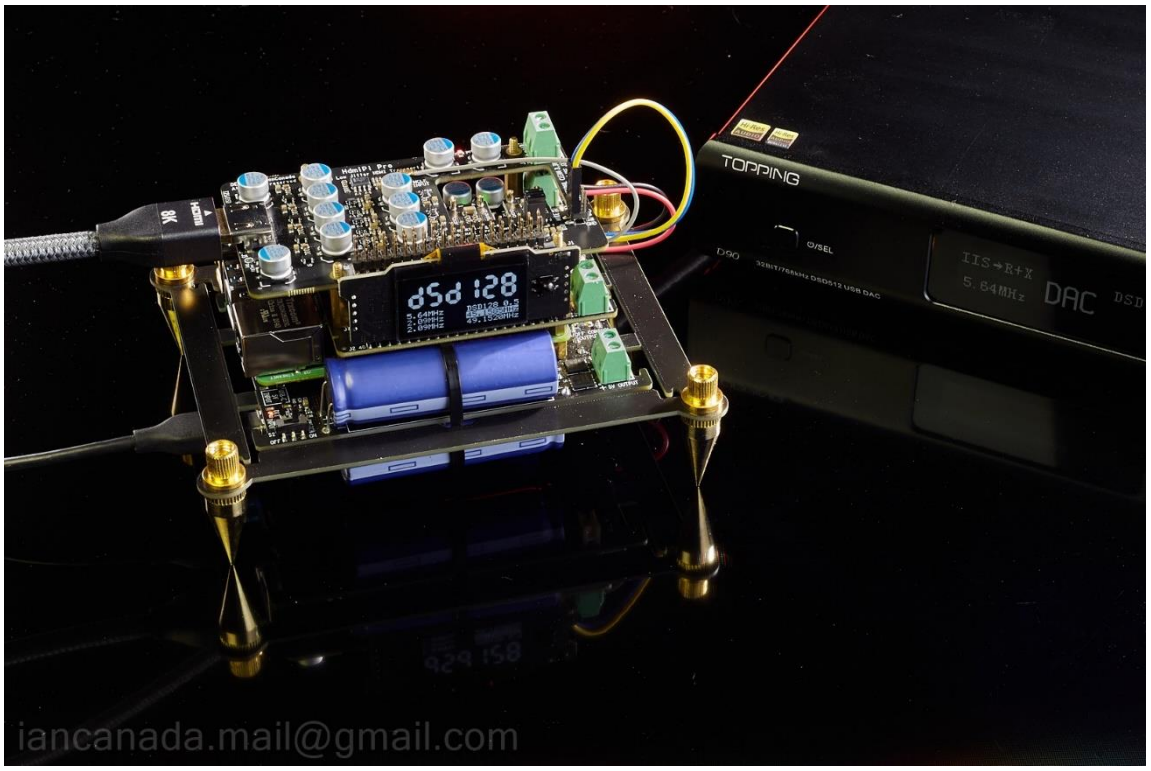
PurePi

Connections

(1). Connect the MCLK from FifoPi to J4 of HdmiPi Pro

(2). Connect the 3.3V pure battery output from J2 of PurePi to 3.3V clean side DC input of FifoPi

(3). Connect DSD/MUTE control signals from FifoPi to J6 of HdmiPi Pro



2. Flagship I2S/DSD over HDMI transport

Components

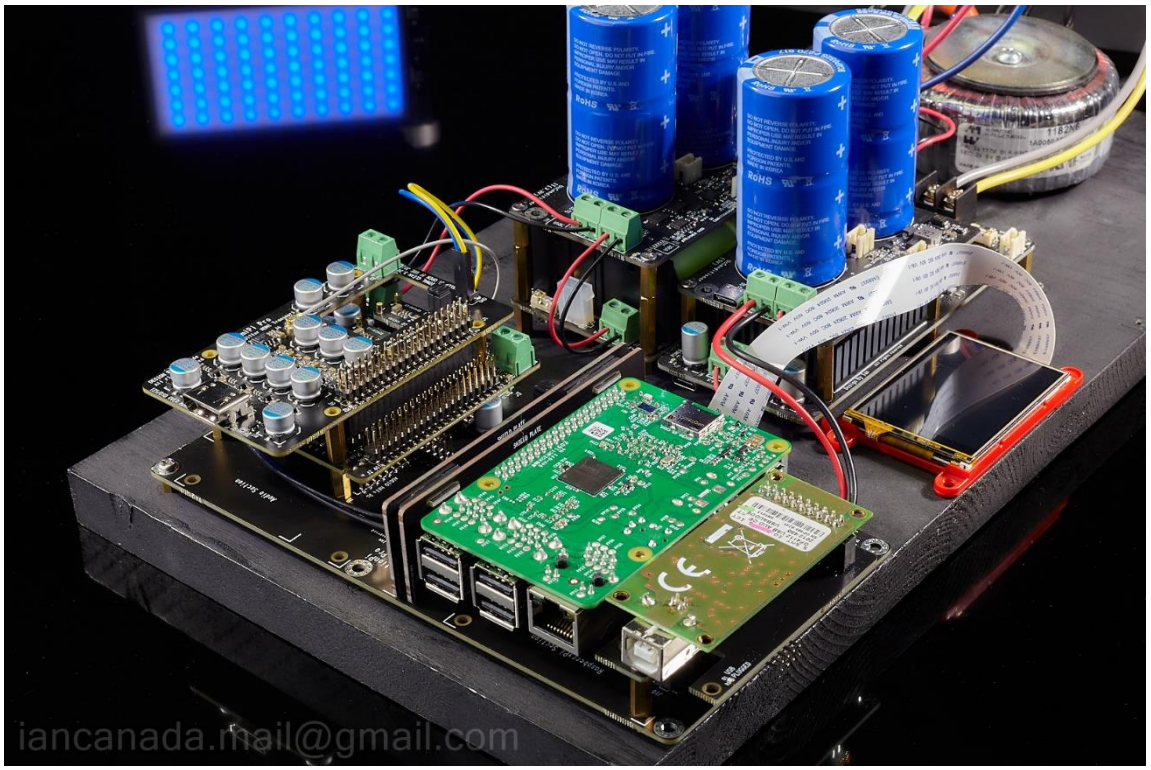
(1). HdmiPi Pro
(2). FifiPi Q7 or FifoPi Q3
(3). StationPi Pro
(4). GEN4-ULCD-24PT Touch screen
(5). Raspberry Pi
(6). Amanero Combo384 or compatible size USB streamer (optional).

Power supply

(1). LinearPi 5V + UcConditioner 5V
(2). LifePO4 Mini + UcConditioner 3.3V
(3). Hammond 1182N6 transformer

Connections

(1). Connect the MCLK from FifoPi to J4 of HdmiPi Pro
(3). Connect DSD/MUTE control signals from FifoPi to J6 of HdmiPi Pro
(3). Connect the 3.3V output from UcConditioner 3.3V to 3.3V clean side DC input of FifoPi
(4). Connect the 5V output from UcConditioner 5V to the 5V input of StationPi pro.
(5). Keep both jumpers at J9 on StationPi Pro short to share 5V with FifoPi Pi side
(6). Connect the control cable from FifoPi to StationPi Pro



J. How to work with external DACs by the I2S/DSD over HDMI input

HdmiPi Pro uses the standard PS Audio format for the I2S/DSD over HDMI connections.

HdmiPi ProII can work with all kinds of DACs even they are not compatible with the standard PS Audio format because of the configurable HDMI format setting switch.

DAC	HdmiPiPro settings	DAC settings
Holo (Spring3 and others)	Default (S1 all OFF for HdmiPi Pro II)	1. I2S Format: ALT2 2. NOS 3. DPLL: OFF
Denafrips (Ares 12th and others)	Default (S1 all OFF for HdmiPi Pro II)	1. I2S Pinout config: MODE0 2. NOS
Topping (D90 or others)	Default (S1 all OFF for HdmiPi Pro II)	1. IIS phase setting: IIS PHASE | REV 2. DSD setting: IIS DSDR | DATA 3. DSD Flag bit: DSD Flag 15
SMSL	Default (S1 all OFF for HdmiPi Pro II)	1. I2S MODE: INVERTED 2. I2S DSD CHANNEL: DSDL=PCM LRCK 3. I2S DSD FLAG=PIN15
Gustard (R26 and others)	S1, HdmiPi Pro II only SW1: OFF SW2: ON SW3: ON SW4: OFF	1. PCM NOS: ON 2. DSD Direct: ON 3. Phase Invert: Disable **Note:** Real test shows that Gustard my reversed the left and right channel internally when play native DSD over HDMI. To play the native DSD correctly, it would be better to setup the FifoPi to DoP pass through mode, make R26 to correct this issue by itself.
MATRIX ELEMENT	S1, HdmiPi Pro II only SW1: OFF SW2: OFF SW3: OFF SW4: ON	Default
Musican	S1, HdmiPi Pro II only SW1: ON SW2: OFF SW3: OFF SW4: OFF	Default

K. Q & A

1. **My DAC has no sound when play DSD music, why?**

   First, you have to make sure your DAC is capable of native DSD music playback. Second, if you DAC cannot automatically switch between PCM and DSD, you have to connect the DSD/MUTE control signals from FifoPi to HdmiPi Pro to enable the DSD mode. Please also make sure the DSD flag signal is connected to PIN15 of the HDMI input of your DAC.

2. **My DAC plays pop sound at the moment of music start/stop, or switch between PCM and DSD. How to fix?**

   You need to connect the DSD/MUTE control signals from FifoPi to HdmiPi Pro as long as your DAC takes MUTE control signal at HDMI PIN16. Or, you can enable the continuous clock mode at FifoPi Q7 to eliminate the pop sound to a best possible degree.

3. **The left and right channels are reversed when I play music, how to solve the problem?**

   It was because the LVDS LRCK+/- signals was opposite to HdmiPi Pro at your DAC input. Please config your DAC's HDMI format to reverse the LRCK+/- signals.

4. **My DAC plays noise rather than music, what's the problem.**

   BCK+/- or DATA+/- signals could be reversed. Please try different HDMI configuration on your DAC to see if the problem can be solved.

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